<Security Level 2>

IEGP

INSPECTION & TEST PLAN FOR INDUCTION MOTORS

DOC. NO. : IEGP-6351-8531-TP-000-1001

PROJECT NAME: PMC SERVICES FOR ETHYLENE GLYCOL PROJECT AT PARADIP REFINERY

TOYO JOB NO. : 6351

OWNER: INDIAN OIL CORPORATION LIMITED.

PMC : TOYO ENGINEERING INDIA PRIVATE LIMITED.

I2	09.02.2018	ISSUED FOR TENDER	RSS	CPM	RPT
I1	20.04.2017	ISSUED AFTER INCORPORATING IOCL COMMENTS	DVB	CPM	RPT
O1	21.11.2016	ISSUED FOR IOCL COMMENTS	DVB	CPM	RPT
REV	**DATE**	**DESCRIPTION**	**MADE BY**	**REVIEWED BY**	**APPROVED BY**

TOYO ENGINEERING INDIA PRIVATE LIMITED
MUMBAI, INDIA

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LV MOTORS

Client : INDIAN OIL CORPORATION LIMITED.	
PMC Job No : 6351	**Applicable Code :**
Item No / Commodity :	**PO. No. :**

Sr. No.		Description of Activity	Inspection By				Certificates / data by Vendor for Approval
			Owner / PMC	Third Party	Detail Engineering Contractor	Vendor	
1		Review of Technical Specification as per PO / Data sheet	H	H	R	P	
2		Detailed Inspection & test plan after issue of PO	A	A	R	P	P
3		Packing & Shipping drawings / documents			A	P	P
4		**Routine Tests :**	H	W	W	P	P
	a	Visual / Dimensional inspection (Including check up of all accessories & name plates, paint shade, Terminal box location(Main, space heater), frame size,earthing terminal,cooling rotation etc.					
	b	Measurement of winding resistance test.					
	c	No load and locked rotor test.					
	d	Insulation Resistance check before and after HV test					
	e	HV & PI test					
	f	Reduced voltage start test.					
5		**Type Tests, if specified**	H	W	W	P	P
	a	Load run test					
	b	Temperature rise test					
	c	Over speed test					
	d	Impulse test					
	e	Vibration test					
	f	Noise level test					
	g	Shaft voltage test					
	h	Overload test					
6		**Special Tests, if specified**	H	W	W	P	P
	a	Critical speed check					
	b	Moment of inertia check					
	c	Tan Delta & polarization index test					

Sr. No.		Description of Activity	Inspection By				Certificates / data by Vendor for Approval
			Owner / PMC	Third Party	Detail Engineering Contractor	Vendor	
	d	Radial air gap measurement check					
7		**Drawing / Documents for Final documentation** :					
	a	Inspection report		R	R	P	P
	b	Inspection release note			P		
	c	Routine test certificates		R	R	P	P
	e	Type test certificates		R	R	P	P
	f	Test certificates for specified area classification, type of enclosure, increased safety/flameproof etc.		R	R	P	P
	g	Calculation of temperature rise,efficiency,PF. Starting torque, starting current, acceleration time etc		R	R	P	P
	h	Manufacturer's Test certificates for bought out items / accessories.		R	R	P	P
	i	Spares & packing list		R	R	P	P
	j	Guarantee certificates		R	R	P	P
	k	"As Built" documents as per PO		R	R	P	P
	l	Installation / Operation / Maintenance manual		R	R	P	P
	m	Catalogues for bought out items / accessories.		R	R	P	P

LEGEND :

H - Hold, A - Approve, P - Prepare / Perform, W - Witness, R - Review

NOTE :

1) Vendor shall submit ITP in line with above and specific technical requirements of purchase order.
2) Vendor shall ensure that all test and measuring instruments are duly calibrated and calibration shall be valid at the time of inspection. Calibration records shall be reviewed by Third Party Inspection Agency/Detail Engineering contractor..

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HV MOTORS

Client : INDIAN OIL CORPORATION LIMITED.	
PMC Job No : 6351	**Applicable Code :**
Item No / Commodity :	**PO. No. :**

Sr. No.		Description of Activity	Inspection By				Certificates / data by Vendor for Approval
			Owner / PMC	Third Party	Detail Engineering Contractor	Vendor	
1		Review of Technical Specification as per PO / Data sheet	H	H	R	P	
2		Detailed Inspection & test plan after issue of PO	A	A	R	P	P
3		Packing & Shipping drawings / documents			A	P	P
4		**Routine Tests :**	H	W	W	P	P
	a	Visual / Dimensional inspection (Including check up of all accessories & name plates, paint shade, Terminal box location (Main, space heater, RTD & BTD), whether main terminal box is segregated or not, details of RTDs, BTDs, frame size, earthing terminal, cooling rotation etc.					
	b	Measurement of winding resistance test.					
	c	No load and locked rotor test.					
	d	Insulation Resistance check before and after HV test					
	e	HV & PI test					
	f	Reduced voltage start test.					
	g	CT polarity,ratio,accuracy class etc. for motors rated above 1000kW (differential protection)					
5		**Type Tests, if specified**	H	W	W	P	P
	a	Load run test					
	b	Temperature rise test					
	c	Over speed test					
	d	Impulse test					
	e	Vibration test					
	f	Noise level test					
	g	Shaft voltage test					
	h	Overload test					

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Sr. No.		Description of Activity	Inspection By				Certificates / data by Vendor for Approval
			Owner / PMC	Third Party	Detail Engineering Contractor	Vendor	
6		**Special Tests, if specified**	H	W	W	P	P
	a	Critical speed check					
	b	Moment of inertia check					
	c	Tan Delta & polarization index test					
	d	Radial air gap measurement check					
7		**Drawing / Documents for Final documentation** :					
	a	Inspection report		R	R	P	P
	b	Inspection release note			P		
	~~b~~ c	Routine test certificates		R	R	P	P
	~~c~~ d	Type test certificates		R	R	P	P
	~~d~~ e	Test certificates for specified area classification, type of enclosure, increased safety/flameproof etc.		R	P	P	R
	~~e~~ f	Calculation of temperature rise,efficiency,PF. Starting torque, starting current, acceleration time etc.		R	P	P	R
	~~d~~ g	Manufacturer's Test certificates for bought out items / accessories.		R	R	P	P
	~~f~~ h	Spares & packing list		R	R	P	P
	~~g~~ i	Guarantee certificates		R	R	P	P
	~~h~~ j	"As Built" documents as per PO		R	R	P	P
	~~i~~ k	Installation / Operation / Maintenance manual		R	R	P	P
	~~j~~ l	Catalogues for bought out items / accessories.		R	R	P	P

LEGEND :

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